Fangjie Gao

+1 734-773-7054 • fangjie@umich.edu

EDUCATION

University of Michigan — **Ann Arbor, MI, United States**
Bachelor's Degree in Industrial and Operations Engineering, Minor in Electrical Engineering — *May 2026*
GPA: 3.82/4.00
Courses: Operations Analysis & Methods, Discrete-Event Simulation, Derivative Instrument, Intro to Data Analytics, Human Ergonomics, Optimization and Computational Methods, Microwave Circuits, Digital Integrated Circuits, Monolithic Amplifier Circuit (audit), Data Structures and Algorithms, Computer Organization

Shanghai Jiao Tong University — **Shanghai, China**
Bachelor's Degree in Electrical and Computer Engineering — *August 2026*
GPA: 3.37/4.00
Courses: Analog Electronic Circuits, Signals and Systems, Electromagnetics, Academic Writing, Linear Algebra, Discrete Mathematics

PROJECT EXPERIENCE

Superheterodyne Receiver of Wi-Fi 5 standard — **Ann Arbor, MI, United States**
EECS411 Microwave Circuit — *August 2025 – December 2025*
- Bandpass input signal into center frequency of 5GHz with bandwidth 160MHz to select useful signal
- Down-convert signal from 5.5GHz to 1.08GHz to 80MHz to support high-speed baseband path and suppress image signal
- Amplify signal with high gain amplifier and low noise amplifier to improve overall gain and signal-to-noise ratio
- Substitute with realistic inductor, capacitor, microstrip line to consider parasitic effects

Low Dropout Voltage Regulator — **Ann Arbor, MI, United States**
EECS413 Monolithic Amplifier Circuit — *August 2025 – December 2025*
- Design in 130-nm process to generate a 1.2 V output from a 1.8 V supply
- Limit quiescent current to be less than 100 uA with load current from 0.1mA to 10mA.
- Implement a PMOS pass transistor architecture with a two-stage amplifier, achieving a low dropout voltage under 0.01V.
- Tune device sizing and use Miller compensation capacitor to achieve phase margin > 89 degrees and power supply rejection ratio of 50dB.
- Pass DRC and LVS for the final layout.

Design a Switched-Capacitor DC-DC Converter — **Shanghai, China**
Advisor: Xuyang Lu — *May 2025 – August 2025*
- Main modules: Charge transfer between 9 capacitors of total 18 phases, realizing a buck converter from 1.8V to 0.9V.
- Control modules: Hysteretic current control using feedback mechanism, stabilizing voltage at any reference level.
- Phase control: Design 18 to 9 bits one-hot decoder with DC and innovus, to decode phases in different cores.

8-bit Full Adder on transistor-level — **Ann Arbor, MI, United States**
EECS312 Digital Integrated Circuit — *January 2025 – April 2025*
- Utilize inversion property of CMOS structure to optimize number of transistors used
- Implement mirror-adder architecture to achieve optimal speed while balancing power trade-offs

Clinic Resource Optimization — **Ann Arbor, MI, United States**
IOE 474 Discrete-Event Simulation — *January 2025 – April 2025*
- Collaborated within a team to build a discrete-event simulation of a clinic using Pro model, analyzing real patient arrival and service data.
- Conducted experiments on combinations of doctor/staffing levels, intake room numbers, and appointment frequencies to optimize resource use and throughput.
- Recommended operational changes that improved resource utilization to 40-70% target range, balanced waiting times, and maximized possible clinic revenue.
- Synthesized findings for faculty in a formal technical paper and presentation.

WORK EXPERIENCE

ECE3120J Digital Integrated Circuit - Shanghai Jiao Tong University — **Shanghai, China**
Teaching Assistant (TA) — *May 2025 - August 2025*
- Grade quizzes, homework, labs and project consistently, resolve students' inquiry, and provide constructive feedback to them.
- Lead lab sections every week to help students learn operations of simulation tool – Virtuoso.
- Coordinate with course instructor and another TA closely to facilitate smooth communication and finish work completely.

Inceptio Technology — **Shanghai, China**
Data Engineer — *August 2023*
- Extract relationships between fuel consumption and other truck factors, such as congestion conditions, wind effects using SQL
- Visualize their relationships to find out factors with significant impacts using Python
- Rank these factors, evaluate each approach's plausibility to reduce fuel consumption.

SKILLS & INTERESTS

Circuit Design: Cadence Virtuoso, Keysight ADS, Synopsys, Innovus, Proteus, Digital
Programming: Python, SQL, C, C++, Verilog, Shell, MATLAB
Computer Tools: SSH, Git, Make, Tmux, Vim, Vivado, Docker
Language: Fluent in English (TOEFL 111). Native in Chinese. | *Hobbies*: Piano, Violin, Swimming, Video Editing, Photography